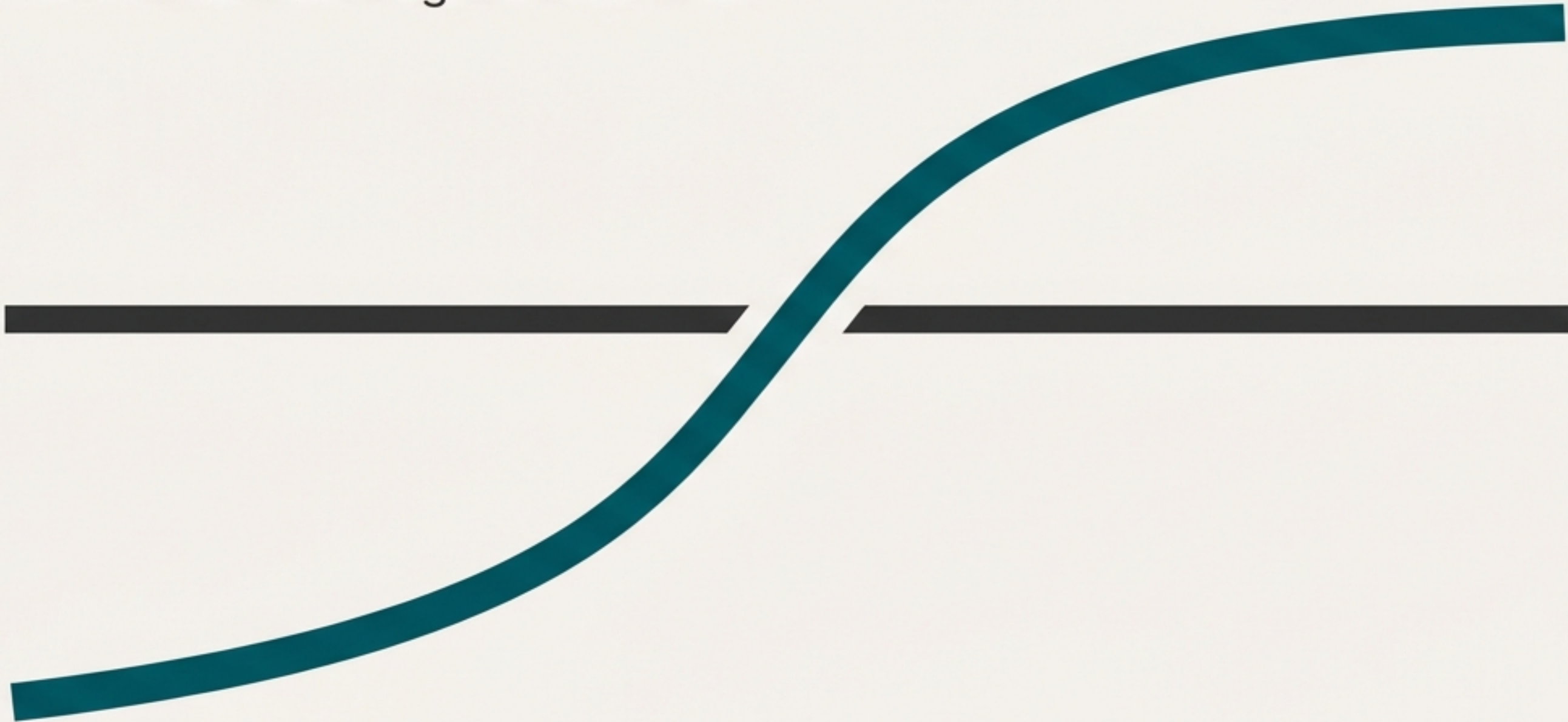


Beyond the Straight Line: Mastering Logistic Regression for Classification

Inter: A storyteller's guide to predicting categorical outcomes and understanding their drivers.



Every day, businesses face a critical question: “Yes or No?”

At its heart, classification is about making a categorical decision based on data. Logistic Regression is a powerful and widely-used tool for answering these questions.



Classification

Will a customer accept a loan offer? Will a transaction be flagged as fraudulent?



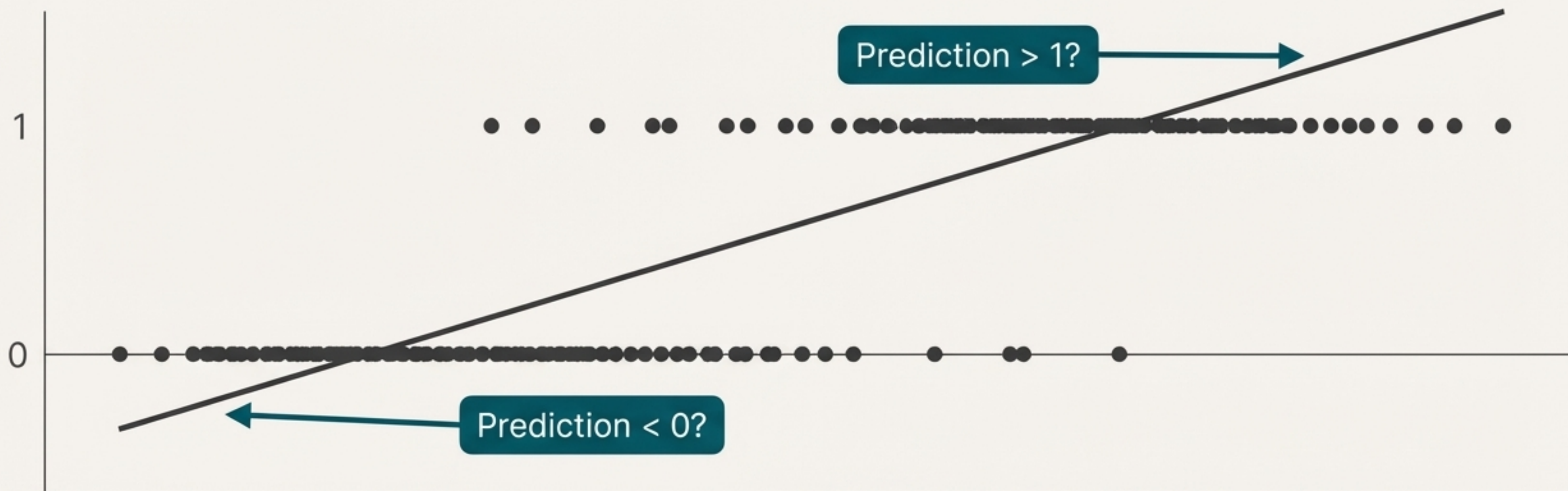
Profiling

What factors differentiate high-value customers from others? What are the characteristics of an employee at risk of attrition?

“Logistic regression can be used for classifying a new record... It can also be used in data where the class is known, to find factors distinguishing between records in different classes...”

Why can't we just use Linear Regression?

While powerful for predicting continuous values (like sales or price), linear regression struggles with binary 0/1 outcomes. When forced to fit this type of data, it can produce nonsensical predictions.

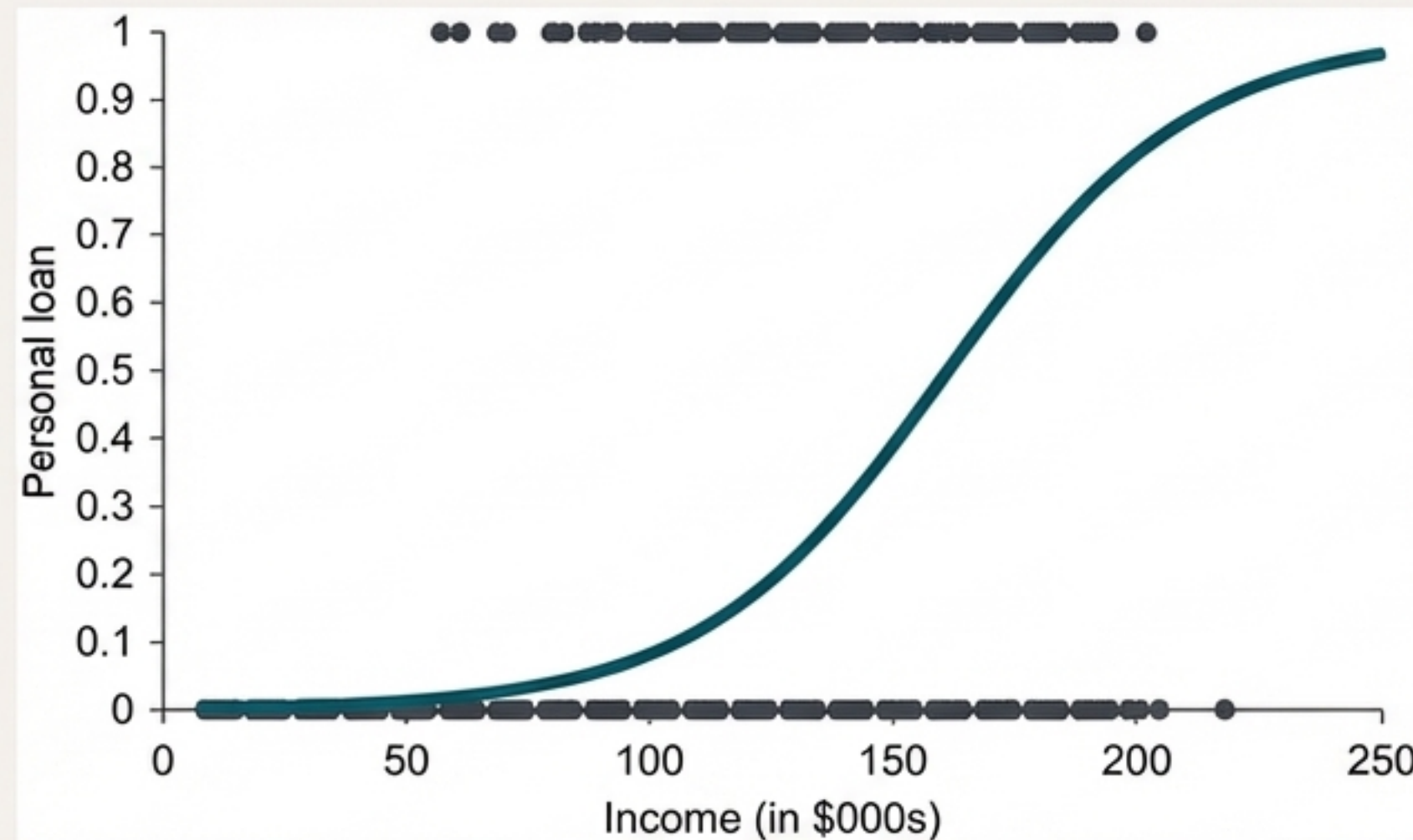


“Linear probability models... do not produce proper predicted probabilities. The numerical predictions they produce are... otherwise meaningless.”

The solution is an **S-shaped** curve that models probability.

Logistic Regression uses a specific non-linear function, the logistic response function, to model the probability of an outcome. This function naturally constrains the output to be between 0 and 1, making it perfect for “Yes/No” predictions.

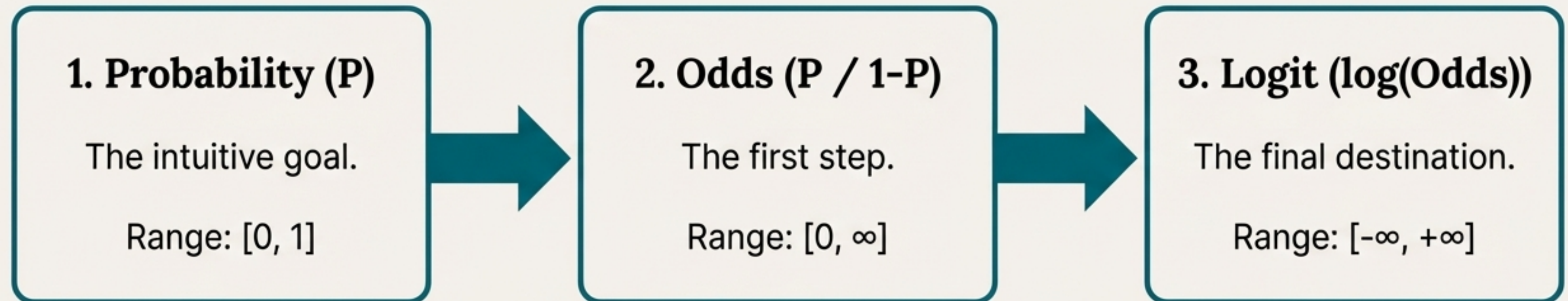
$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_q x_q)}}$$



The S-shaped curve plots the probability of a customer accepting a loan based on their income. Notice how it never goes above 1 or below 0.

To build this curve, we need a clever transformation.

The key idea is to transform our target variable, Y , into something that isn't bounded between 0 and 1. This allows us to use a linear combination of predictors, just like in linear regression. This is a three-step journey.



By modeling the logit, we can relate our predictors to the outcome with a simple linear equation.

The Engine Room: From Probability to Logit.

Let's walk through how these three concepts relate to each other. Each step gets us closer to a variable that can be modeled linearly.

Probability (P)

What it is: The probability of success ($Y=1$).

Range: $[0, 1]$.

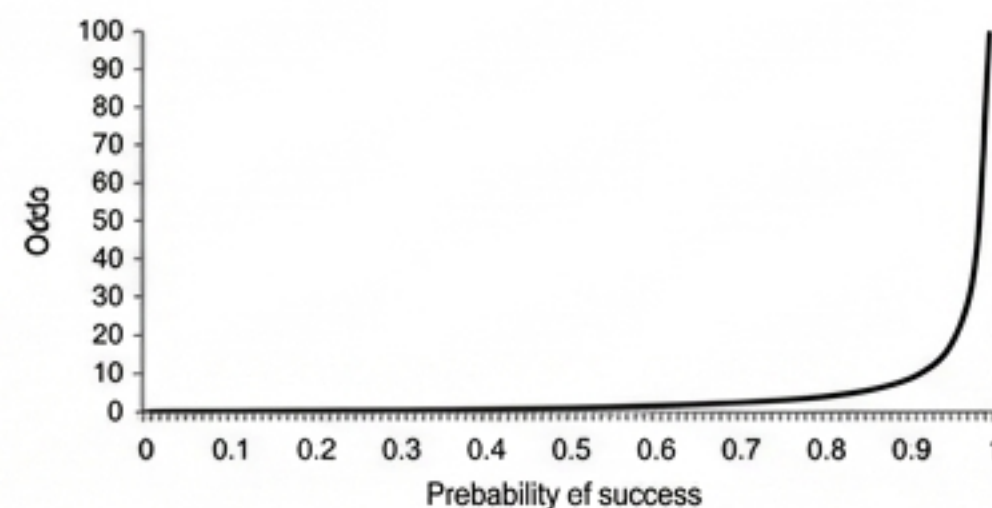
Challenge: The boundaries at 0 and 1 make it hard to model with a linear function.

Odds ($P / 1-P$)

What it is: The ratio of the probability of success to the probability of failure.

Range: $[0, \infty]$.

Progress: Solves the upper boundary problem, but still bounded at 0.

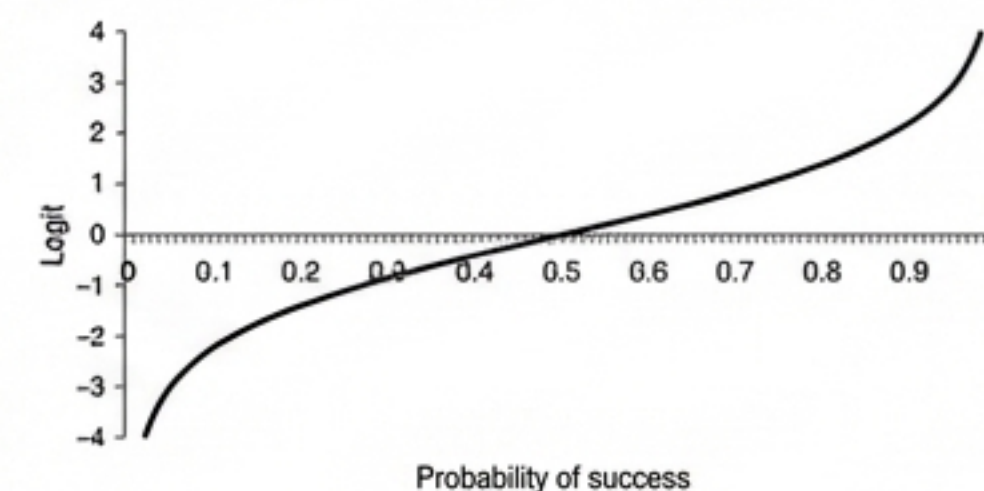


Logit ($\log(\text{Odds})$)

What it is: The natural logarithm of the odds.

Range: $[-\infty, +\infty]$.

The "Aha!" Moment: The logit is unbounded and can be modeled as a linear function of the predictors!



The Final Equation: $\log(\text{odds}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_q x_q$

Case Study: Who will accept a personal loan from Universal Bank?

The Business Challenge

Universal Bank wants to improve the efficiency of its marketing campaigns by identifying customers most likely to accept a personal loan.

The Data

- **Dataset:** 5,000 customers.
- **Outcome:** Only 480 customers (9.6%) accepted the loan in a previous campaign.
- **Goal:** Build a logistic regression model to predict acceptance for future campaigns.

Predictors at a Glance

- **Demographics:**
 - Age
 - Family Size
- **Financials:**
 - Income
 - CCAvg (Avg. credit card spending)
 - Mortgage
- **Bank Relationship:**
 - Securities Account
 - CD Account
 - Online Banking
 - Credit Card

The model reveals powerful drivers of loan acceptance.

The Estimated Model Equation:

$$\text{Logit}(\text{Loan} = \text{Yes}) = -12.293 + 0.060 * \text{Income} + 0.606 * \text{Family} + \dots + 3.514 * \text{CD Account} + \dots$$

Interpreting the Results in Terms of Odds:

We can interpret the coefficients by taking their exponent (e^β). This tells us the multiplicative factor by which the odds of acceptance change for a one-unit increase in the predictor.

Key Insight: Income

For each additional \$1,000 in annual income, the odds of a customer accepting the loan increase by a factor of $e^{0.060}$ (approx. **6.2%**), holding other variables constant.

Key Insight: CD Account (The Game Changer)

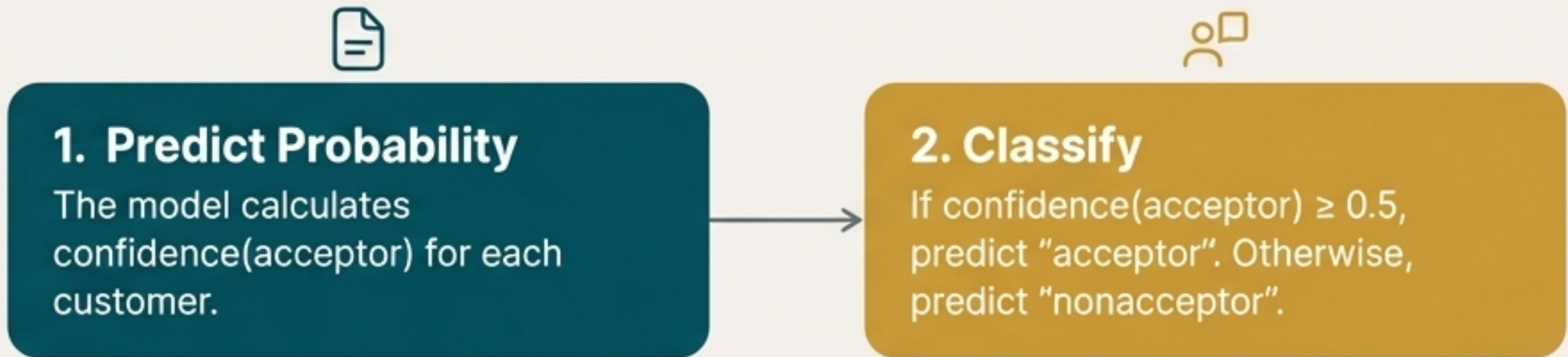
The coefficient for having a CD Account is 3.514. This means the odds of acceptance for a customer with a CD account are **$e^{3.514} = 33.58$ times higher** than for a customer without one. This is a massive signal for targeting.

Visual Anchor

Attribute	Coefficient
Income	0.060
CD Account	3.514

From probability to a clear decision: Applying the threshold

The logistic regression model outputs a probability (or “propensity”) for each customer. To classify them as an “acceptor” or “nonacceptor,” we apply a decision threshold. A common starting point is 0.5.



Example in Action:

The table below shows the model's output for five customers. For Row 4, the model was confident the customer would accept (probability = 0.579), but the actual outcome was “nonacceptor”—a misclassification.

Row No.	ID	Personal Loan	prediction(Personal Loan)	confidence(nonacceptor)	confidence(acceptor)	Education	Experience	Age	Income	Family	CCAvg	Mortgage
1	6	nonacceptor	nonacceptor	0.996	0.004	2	13	37	29	4	0.400	155
2	7	nonacceptor	nonacceptor	0.980	0.020	2	27	53	72	2	1.500	0
3	9	nonacceptor	nonacceptor	0.953	0.047	2	10	35	81	3	0.600	104
4	11	nonacceptor	acceptor	0.421	0.579	3	39	65	105	4	2.400	0
5	13	nonacceptor	nonacceptor	0.824	0.176	3	23	48	114	2	3.800	0

How well did our model perform? The Confusion Matrix.

To evaluate the model, we use a holdout dataset (data the model has not seen before) and compare its predictions to the actual outcomes. The confusion matrix summarizes this performance.

Overall Accuracy: 96.05% -
The percentage of total correct predictions.

	<u>true nonacceptor</u>	<u>true acceptor</u>
<u>pred. nonacceptor</u>	1791	62
<u>pred. acceptor</u>	17	130

Recall (Nonacceptors): 99.06% - The model is excellent at correctly identifying those who will *not* accept the loan.

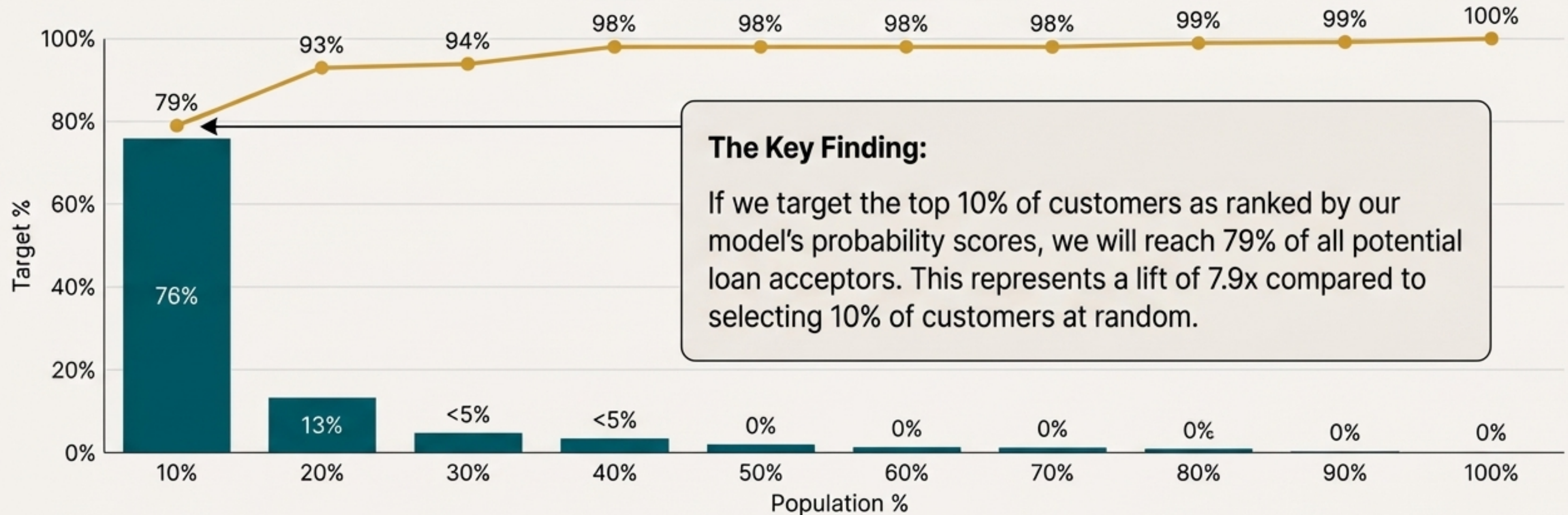
Recall (Acceptors): 67.71% - We correctly identified 130 out of the 192 actual acceptors. This means we missed about one-third of them.

Takeaway: The model is highly accurate overall, but its ability to find acceptors could be improved, perhaps by adjusting the classification threshold.

Beyond Accuracy: The business value is in the lift.

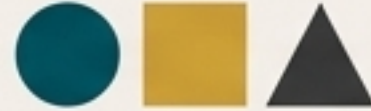
For a marketing campaign, we don't necessarily need to classify everyone perfectly. We need to efficiently find the customers most likely to say "yes". The lift chart measures exactly this: how much better our model is than random guessing.

The Lift Chart for Universal Bank:



This demonstrates immense business value. The model allows for highly targeted, efficient marketing campaigns.

What if you have more than two choices?



The principles of logistic regression can be extended to handle multi-class classification problems, where the outcome has three or more nominal categories.

The Core Idea: One class is chosen as a “reference” category. The model then builds a set of equations that predict the log-odds of a record belonging to each other class relative to the reference. A record is ultimately assigned to the class with the highest predicted probability.

Example: Accident Severity Prediction

Goal: Predict accident severity based on factors like weather and alcohol involvement.

Classes: (0) No Injury, (1) Non-fatal Injury, (2) Fatality

Method: Using "No Injury" as the reference, the model would predict $\log[P(\text{Injury})/P(\text{No Injury})]$ and $\log[P(\text{Fatality})/P(\text{No Injury})]$.

Logistic Regression: Your Toolkit for Classification and Profiling

A powerful, interpretable, and efficient method for modeling categorical outcomes.

The Problem It Solves

Predicts binary (Yes/No) or multi-class outcomes where linear regression fails.

How It Works

Models the **logit (log-odds)** of an event as a linear function of predictors, then maps this back to a probability via the logistic function (S-curve).

Its Dual Power

- **Classification**
Assigns new records to a class based on predicted probability and a threshold. Performance is measured by confusion matrices and lift charts.
- **Profiling**
Interprets model coefficients (e^{β}) to understand the magnitude and direction of each predictor's influence on the outcome.